Project Management Plan: Software Tracking Database Aaron McCullough

Introduction

In a large-scale academic environment, such as a major research university, the volume of software assets, ranging from site-licensed operating systems to specialized departmental research tools, requires a robust management system. Currently, many institutions face challenges in tracking installation dates, licensing compliance, and version control across various departments. This project aims to develop a centralized software-tracking database for the IT Department. This database will serve as the single source of truth for software metadata, including developer information, license types (e.g., seat-based, concurrent, or site-wide), and the specific hardware on which the software is deployed. By implementing this system, the institution will mitigate legal risks associated with licensing audits and optimize its procurement spending.

Phase 1: Initiation Phase

The primary goal of the initiation phase is to determine the feasibility of the project and secure formal approval. During this stage, the Project Manager (PM) will meet with the Chief Information Officer (CIO) and the IT Director to define the project's boundaries. The main "deliverable" of this phase is a Project Proposal or Charter.

- **Activities:** Conducting a needs assessment to identify gaps in current tracking methods and performing a cost-benefit analysis of building a custom database versus purchasing a commercial off-the-shelf solution.
- **Responsible Employees:** Project Manager, IT Director, and a Financial Analyst.
- **Stakeholders:** IT Administration and the University Procurement Office.
- **Decision-making:** A "Go/No-Go" decision will be made by the Project Board based on the estimated ROI and resource availability.

Phase 2: Definition Phase

Once the project is approved, the definition phase focuses on gathering the specific requirements. According to Baars (2006), this phase involves determining what the project results must meet. This is critical for an educational setting where software requirements vary significantly between the Arts and Engineering departments.

- **Activities:** Interviewing department heads to identify mandatory data fields (e.g., "installation date," "licensing agreement type") and determining user access levels.
- **Responsible Employees:** Project Manager and Business Analysts.
- **Stakeholders:** Departmental IT Liaisons and Faculty representatives.
- **Deliverables:** A Functional Requirements Document (FRD) and a Project Result Description.

- **Assessment:** Stakeholders will sign off on the requirements to prevent "scope creep" during later stages.

Phase 3: Design Phase

The design phase translates the requirements into a technical blueprint. For a tracking database, this involves designing the data schema to ensure that a single software version can be linked to multiple licenses and multiple physical machines.

- **Activities:** Creating Entity-Relationship Diagrams (ERDs), designing user interface (UI) wireframes for IT staff, and selecting the technology stack (e.g., SQL-based architecture).
- **Responsible Employees:** Database Architect and UI/UX Designer.
- **Deliverables:** A Technical Design Document and a clickable prototype of the user interface.
- **Testing/Piloting:** The PM will conduct a "design walk-through" with the IT technicians who will be the primary users to ensure the workflow is intuitive.

Phase 4: Development Phase

During the development phase, the actual construction of the database occurs. The project team builds the backend architecture and the frontend interface based on the designs approved in the previous phase.

- **Activities:** Database table creation, coding the search and reporting functionalities, and integrating the database with existing university hardware inventory systems.
- **Responsible Employees:** Software Developers and Database Administrators (DBAs).
- **Deliverables:** A fully functional beta version of the software-tracking database.
- **Testing:** Alpha testing is conducted by the development team to identify bugs, followed by internal IT unit testing to ensure data integrity.

Phase 5: Implementation Phase

The implementation phase involves the actual deployment of the database into the university's live environment. This is often the most sensitive phase, as it requires moving data from legacy spreadsheets into the new system.

- **Activities:** Data migration, installing the software on IT servers, and conducting training sessions for departmental IT coordinators.
- **Responsible Employees:** Implementation Specialist, Project Manager, and IT Support Staff.
- **Stakeholders:** All university departments that must report their software usage.
- **Pilot:** A "soft launch" will occur within the College of Arts and Sciences to identify any real-world performance issues before a university-wide rollout.

Phase 6: Follow-up Phase

The final phase focuses on the long-term sustainability of the project and the formal closure of the development cycle. As Baars (2006) notes, this phase ensures that the project results are maintained and that the team reflects on the process.

- **Activities:** Setting up a maintenance schedule, creating a "Help Desk" protocol for database issues, and conducting a Post-Implementation Review (PIR).
- **Responsible Employees:** Project Manager and IT Maintenance Team.
- **Deliverables:** Final Project Report, User Manuals, and a Project Evaluation.
- **Assessment:** Success is measured against the original requirements, specifically, whether the IT department can successfully generate a licensing compliance report within five minutes.

Conclusion

Developing a software-tracking database for a university is a complex undertaking that requires strict adherence to project management principles to avoid budget overruns and technical failure. By following the six phases, Initiation, Definition, Design, Development, Implementation, and Follow-up, the project team can ensure that the final product meets the needs of all academic departments while providing the IT department with the tools necessary for effective institutional oversight. This structured approach provides the transparency needed for high-stakes decisions regarding the university's digital assets.

References

Baars, W. (2006). *Project Management Handbook* (Version 1.1). Data Archiving and Networked Services (DANS). https://www.projectmanagement-training.net/wp-content/uploads/2016/08/project_management_handbook.pdf